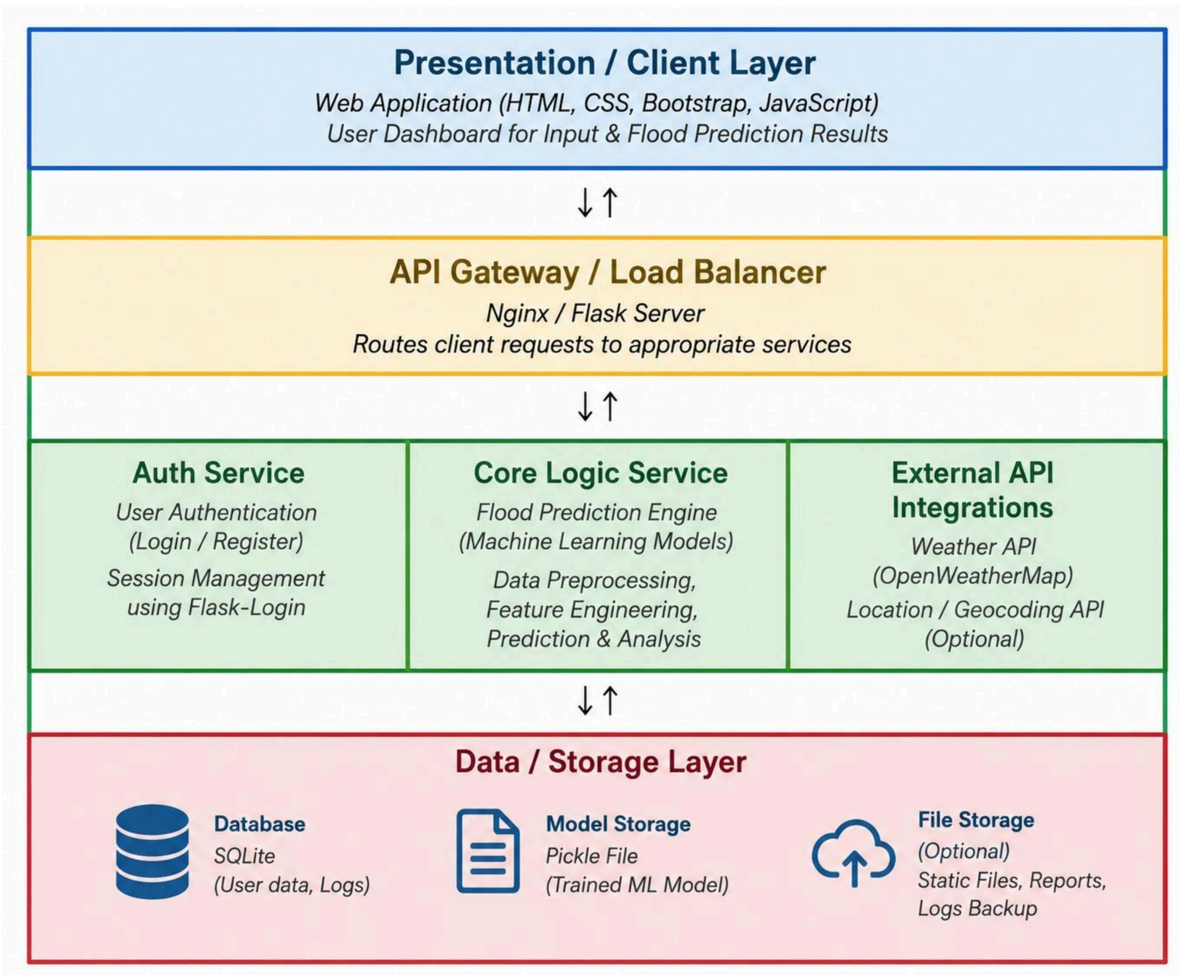
Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter weather parameters and view flood prediction results.	HTML, CSS, JavaScript, Flask Templates
API Gateway/Application Layer	Handles user requests, processes form inputs, validates weather parameters, and connects the frontend with the machine learning prediction model.	Flask (Python)
Authentication Module	Provides direct access to the prediction system without authentication or role-based login.	Flask Templates, HTML, CSS, JavaScript
Core Logic Service	Performs data preprocessing, feature scaling, loads the trained XGBoost model, and predicts flood probability.	Python, Scikit-learn, XGBoost, NumPy, Pandas, Joblib
External Data Integration	Uses the flood dataset for machine learning model training and evaluation. The Flask application is deployed using Render Cloud Platform.	Dataset File, Render Cloud Platform
Microservices	Performs data preprocessing, applies feature scaling, loads the trained XGBoost model, and generates flood risk predictions based on user input.	XGBoost, Scikit-learn, StandardScaler, Joblib
Database	Stores the training dataset and saved machine learning files required for prediction. No database is used in the current implementation.	Dataset File, Joblib Model Storage (floods.save, transform.save)